WEEKLY PROGRESS REPORT

WEEK 04: EMBEDDED SYSTEM TRAINING — CIRCUIT DESIGN & I/O FUNDAMENTALS

Intern Name: Gayatri **Week:** Week 04

Intern Domain: IoT / Embedded Systems **Training Focus:** Embedded System Training — Circuit Design

Weekly Focus	Basic Electronics, Input/Output Systems, Sensors & Actuators, ADC/DAC, Circuit Design Concepts
Assessment	IoT Test 2 and IoT internship assignment
Report Type	Weekly Progress Report — Week 04

I. OVERVIEW

This week focused on **Embedded System Training with an emphasis on circuit design, electronic components, input-output systems, and fundamental IoT concepts**. The training was organized into three major parts. The first part established a foundation in basic electronics and electrical measurements. The second part introduced input and output concepts, sensors, actuators, signal conversion, and process control. The third part focused on circuit designing concepts and important design parameters used in electronic systems.

Along with the technical learning, I completed the related **IoT internship assignment** and participated in **IoT Test 2**. The week's activities helped connect theoretical electronic concepts with the way components, signals, and circuits are used in embedded and IoT systems.

II. ACTIVITIES & ACHIEVEMENTS

1. Basic Electronics and Electrical Fundamentals

The first section of the training introduced the basic building blocks of electricity and electronics. I studied the fundamental electrical quantities and their measurement, followed by circuit representation and the working concepts of commonly used electronic components.

- **Electricity fundamentals:** introduction to the basic components and concepts of electricity.
- **Volt, Ohm and meter basics:** understanding basic electrical quantities and the role of meters in measuring electrical values.
- **Circuit diagrams:** learning the basics of circuit diagrams and electronic roadmaps used to represent circuits.
- **Resistors:** understanding their role in controlling current and their use in electronic circuits.
- **Ohm's Law:** studying the relationship between voltage, current and resistance.
- **Capacitors:** learning the basic purpose and behavior of capacitors in circuits.
- **Inductors:** understanding the basic role of inductive components.
- **Diodes:** studying the basic function of diodes as electronic components.
- **Transistors / electronic valves:** learning the introductory concept and role of transistors in electronic systems.

2. Input, Output and Signal Conversion

The second section expanded the learning from individual components to the interaction between an embedded system and its surrounding environment. I studied how systems receive information, process it, and produce outputs.

- **Input and output concepts:** understanding how information enters and leaves an electronic or embedded system.
- **Basic electronic knowledge:** reinforcing the electronic principles required for working with embedded systems.
- **Sensors and actuators:** studying sensors as input elements and actuators as output/control elements.
- **Analog-to-Digital Conversion (ADC):** learning the basic conversion of analog signals into digital data.
- **Digital-to-Analog Conversion (DAC):** understanding the reverse process of producing analog signals from digital information.
- **Discrete-data I/O devices:** studying input-output devices that work with discrete digital states.
- **Process control systems:** understanding the basic relationship between sensing, processing, control, and output.

3. Circuit Designing Concepts

The third section focused on circuit design and the factors that influence how electronic circuits behave. The training covered design abstraction, design terminology, design concepts and commonly used circuit-design parameters.

- **Design abstraction level:** understanding how a circuit can be represented at different levels of detail.
- **Design concepts and metrics:** studying the factors used to describe and evaluate circuit designs.
- **Circuit-design terminology:** becoming familiar with important terms used in electronic design.
- **Impedance matching:** understanding its importance in transferring signals appropriately between circuit sections.
- **BCB/related circuit-design concepts:** studying the terminology and concepts included in the training material.
- **Noise margin:** understanding the tolerance of a digital circuit to unwanted noise while maintaining correct logic levels.
- **Pull-up and pull-down resistors:** learning how resistors can establish defined logic states at digital inputs.
- **Filters:** studying the basic purpose of filtering unwanted signal components.
- **Loading effect:** understanding how connecting one circuit stage to another can affect circuit behavior.
- **Fan-in and fan-out:** learning how digital circuit inputs and outputs relate to multiple logic connections.

4. IoT Internship Assignment and Test 2

After completing the relevant training topics, I worked on the **IoT internship assignment** and completed **IoT Test 2**. These activities provided an opportunity to review the concepts studied during the week and check my understanding through assessment.

III. CHALLENGES AND HURDLES

The main challenge during this week's training was understanding how individual electronics concepts connect together within a complete embedded system. Several topics required moving from component-level understanding to system-level thinking.

- **Connecting theory with circuits:** Understanding how resistors, capacitors, diodes, inductors and transistors contribute to an actual circuit required careful review of their individual roles.
- **Understanding signal conversion:** ADC and DAC concepts required distinguishing between continuous analog information and discrete digital representation.
- **Understanding input-output relationships:** Sensors, actuators and process-control systems required thinking about the complete flow from input to processing and output.
- **Remembering circuit-design parameters:** Concepts such as impedance matching, noise margin, pull-up/pull-down resistors, filters, loading effect, fan-in and fan-out introduced several new technical terms.
- **Assessment preparation:** Reviewing the large number of concepts before the IoT Test 2 required organizing the topics and revising their practical purpose rather than relying only on memorization.

Approach Used to Handle the Challenges

- I revised the topics in the same three-part structure used in the training: **basic electronics** → **input/output systems** → **circuit design**.
- I focused on understanding the **purpose and role** of each component or design concept instead of memorizing definitions alone.
- I connected sensors and actuators with the broader **input-process-output** flow of embedded and IoT systems.
- I used quizzes, assignment work and test preparation as checkpoints to identify topics that needed additional revision.

IV. LESSONS LEARNED

- **Electronics is the foundation of embedded systems:** I learned that understanding basic electrical quantities and components is important before working with more advanced embedded concepts.
- **Ohm's Law is a fundamental circuit relationship:** I strengthened my understanding of the relationship among voltage, current and resistance.
- **Components have specific roles:** Resistors, capacitors, inductors, diodes and transistors perform different functions and together form useful electronic circuits.
- **Embedded systems interact with the real world:** Sensors provide information from the environment while actuators allow a system to produce physical or controlled outputs.
- **Analog and digital signals are different:** I learned the basic purpose of ADC and DAC in connecting analog-world signals with digital systems.
- **Circuit design involves trade-offs and system behavior:** Parameters such as impedance matching, noise margin, loading effect, fan-in and fan-out help determine how reliably circuit sections operate together.
- **Defined digital states are important:** Pull-up and pull-down resistors help avoid undefined input states in appropriate digital circuits.
- **Assessment improves revision:** Completing the IoT assignment and IoT Test 2 helped me identify how well I could apply the concepts covered during the training.

V. WEEKLY LEARNING SUMMARY

Training Part	Major Topics Covered	Outcome
Part 1 — Basic Electronics	Electricity, volt/ohm/meter basics, measuring electricity, circuit diagrams, resistors, Ohm's Law, capacitors, inductors, diodes, transistors.	Built a foundation in electronic components and basic circuit concepts.
Part 2 — Input & Output	I/O, sensors, actuators, ADC, DAC, discrete-data I/O devices, process control systems, quizzes.	Understood how embedded systems receive, convert and produce information.
Part 3 — Circuit Design	Design abstraction, design concepts/metrics, terminology, impedance matching, noise margin, pull-up/down resistors, filters, loading effect, fan-in/fan-out.	Developed introductory understanding of circuit design considerations.
Assessment	IoT internship assignment and IoT Test 2.	Reviewed and assessed the week's learning.

VI. NEXT WEEK'S GOALS

- Continue strengthening the fundamentals of **embedded systems and IoT** through practical examples.
- Revise circuit components and design parameters so that their roles can be applied confidently in practical circuit problems.
- Practice identifying appropriate **inputs, outputs, sensors and actuators** for different IoT use cases.
- Apply the understanding of electronics and circuit design to upcoming internship/project activities.
- Review assessment feedback and revisit any topics that were difficult during **IoT Test 2**.

VII. ADDITIONAL COMMENTS

This week provided a broad and useful foundation in embedded-system electronics. Moving from basic electrical components to input-output systems and then to circuit-design considerations helped me understand how different concepts fit together in an IoT environment. Completing the internship assignment and IoT Test 2 also gave me an opportunity to revise and assess my learning. I will continue building on these fundamentals through practical embedded-system and IoT work in the coming weeks.

Submitted by: Gayatri

Domain: IoT / Embedded Systems

Report: Week 04